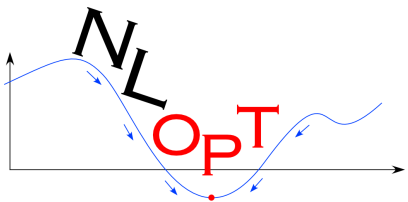




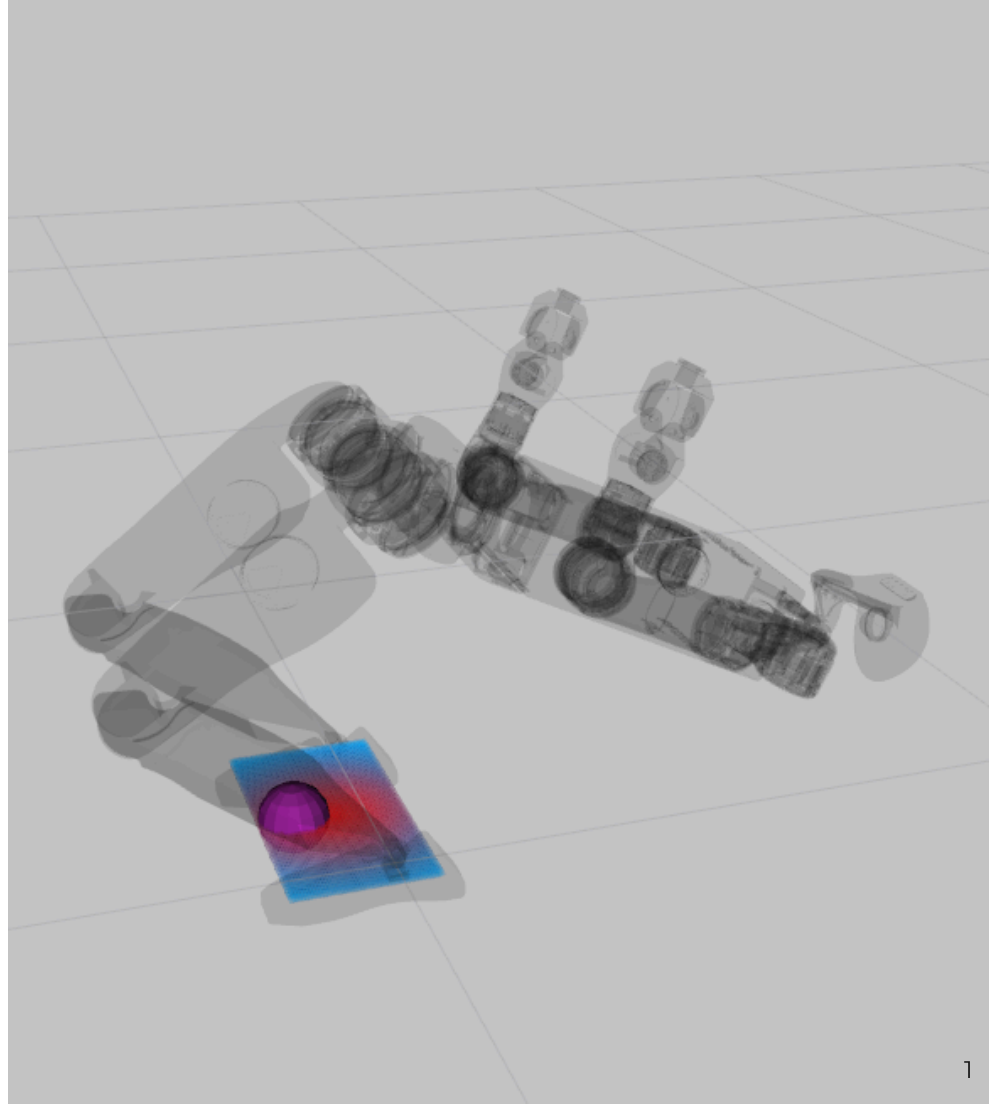
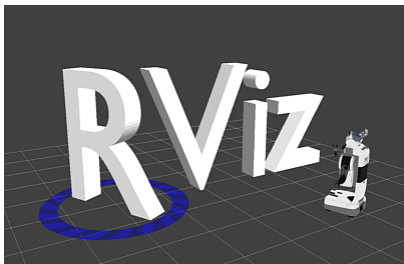
# Project Neo - RELOADED

Dominic Zahn, 3/25/2026

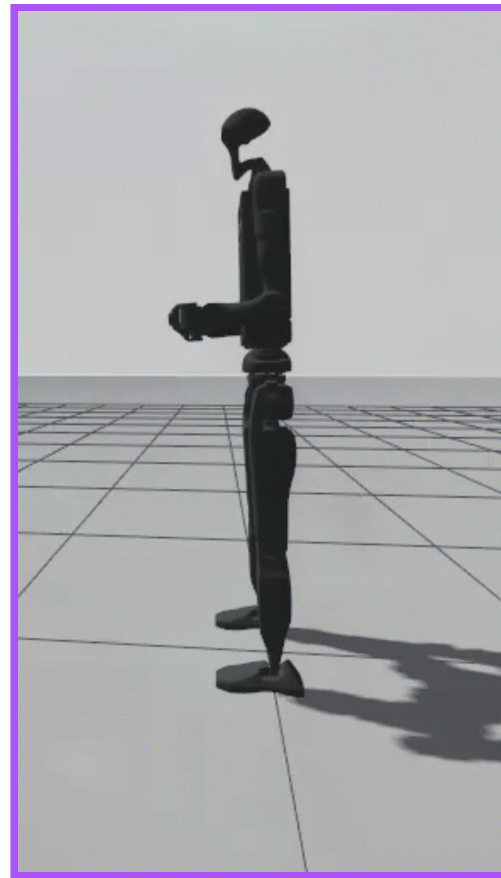
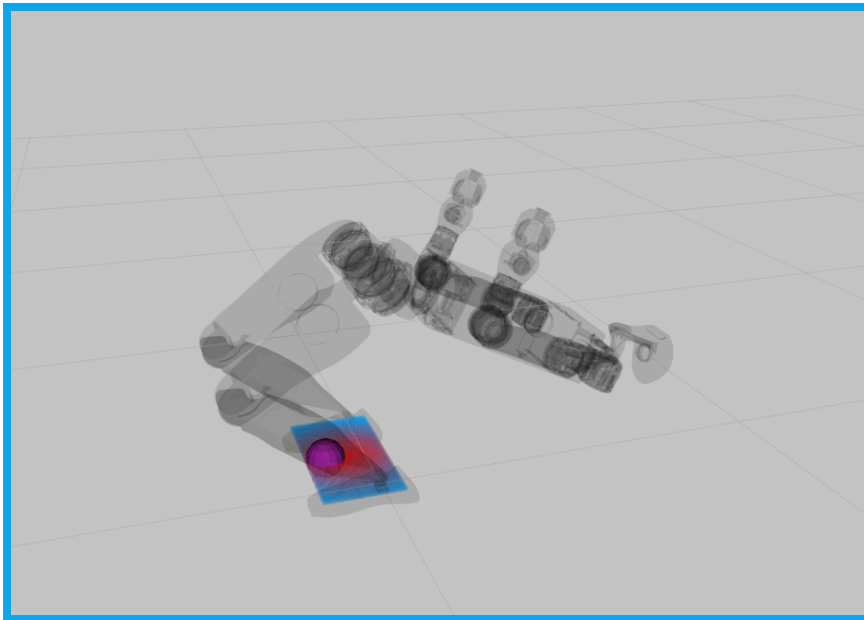
what happened before ... 



 Static Pose Optimization



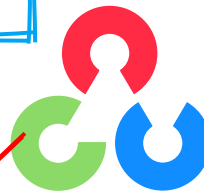
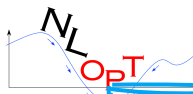
# From Pose to Motion



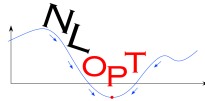
# Previous Tech Stack



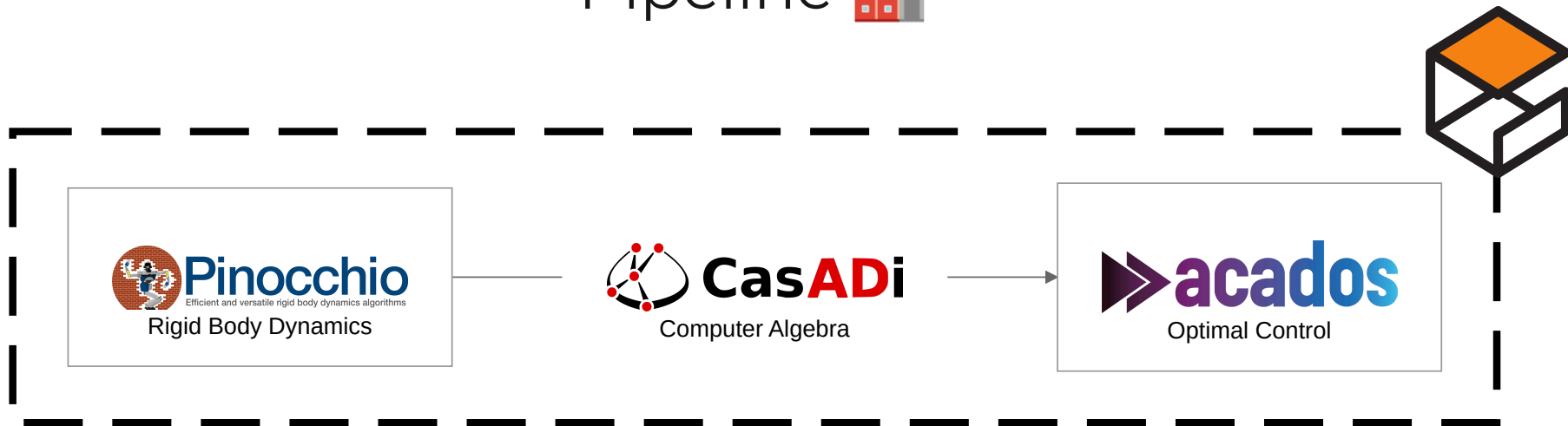
RBDL



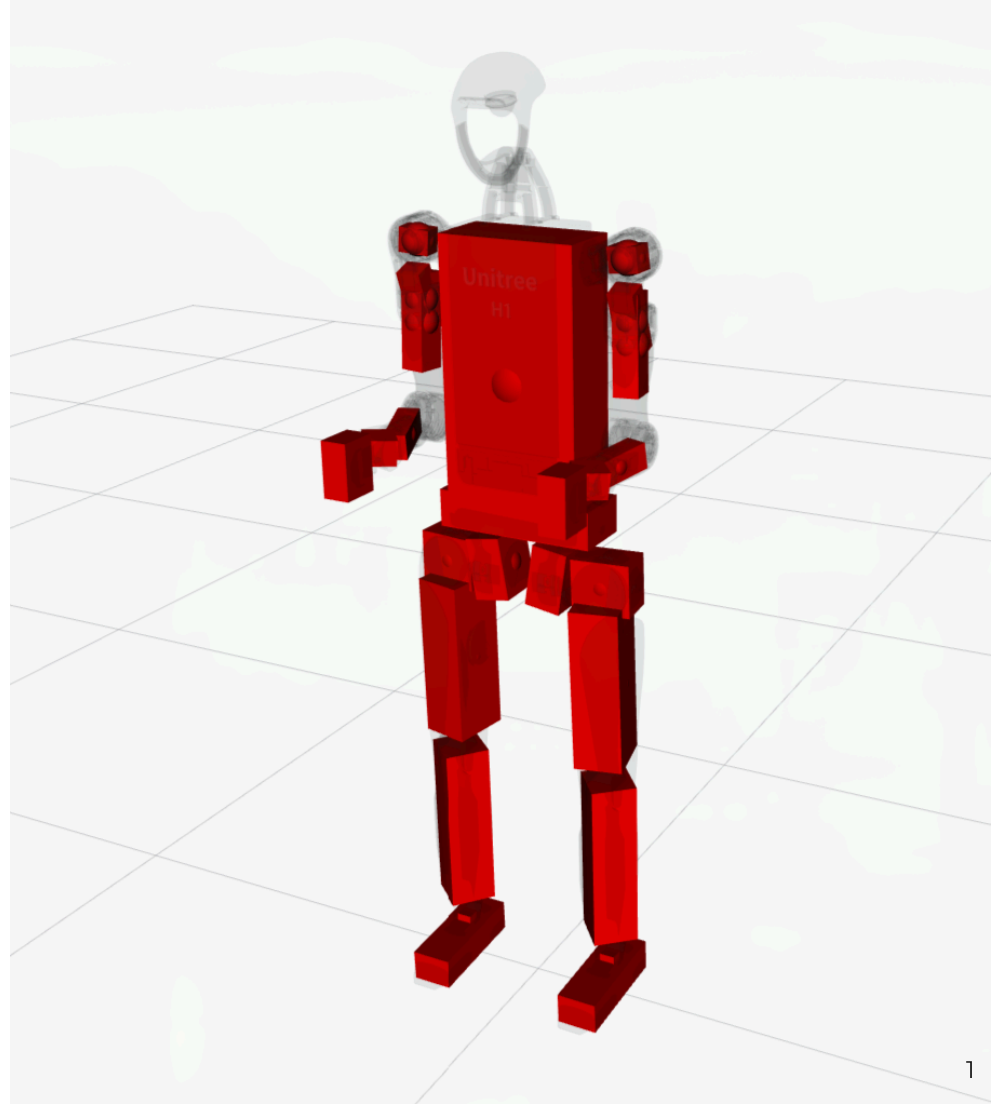
# Updated Control Tech Stack



# Pipeline

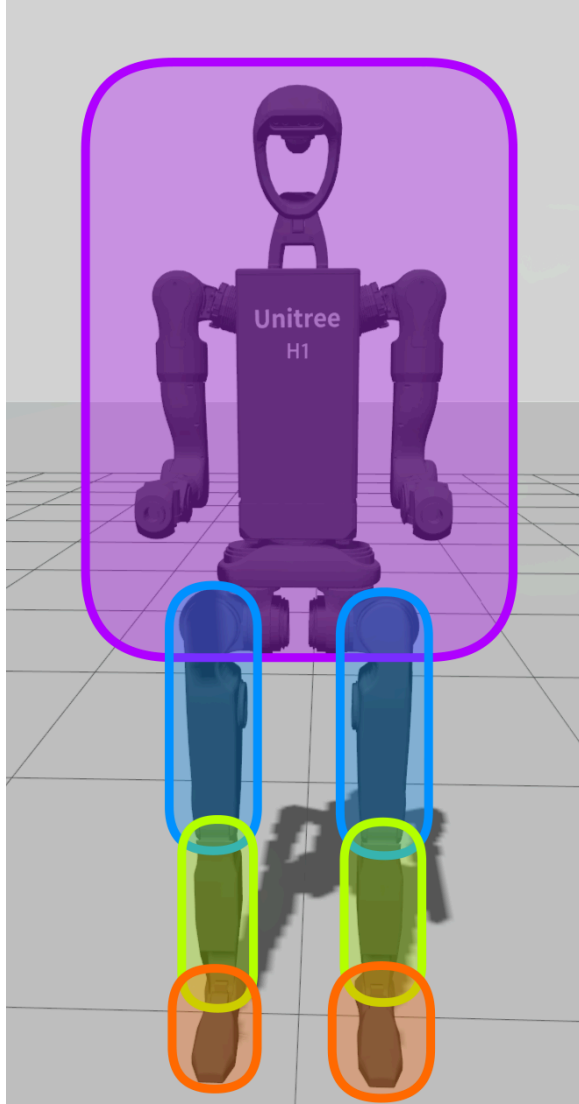
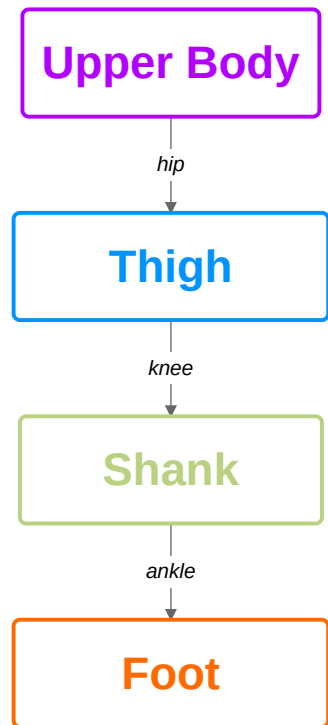


How to Model  
Heinz? 🧬

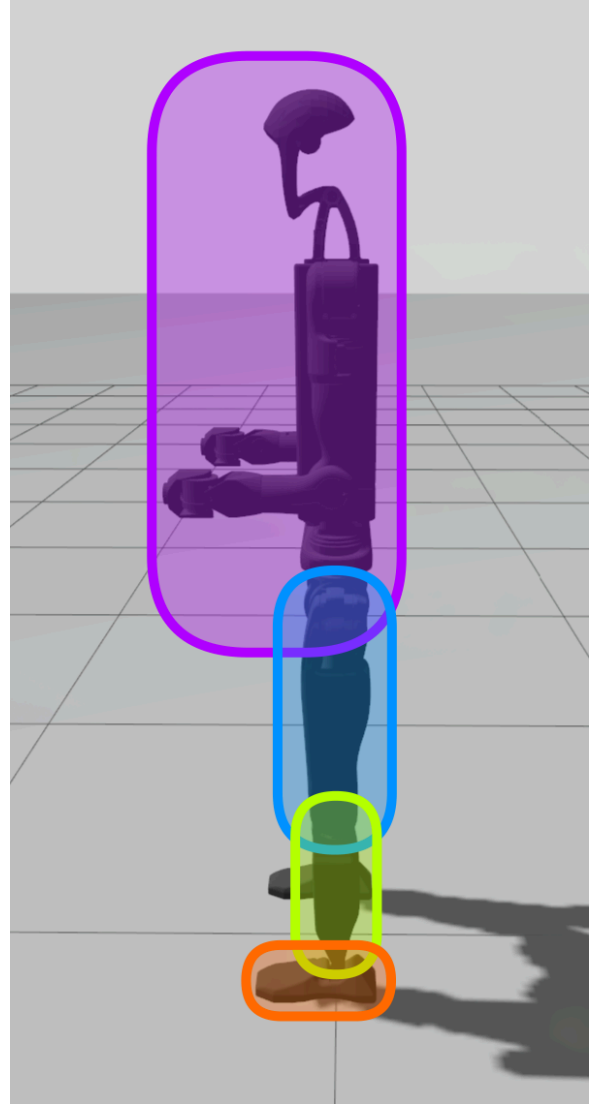
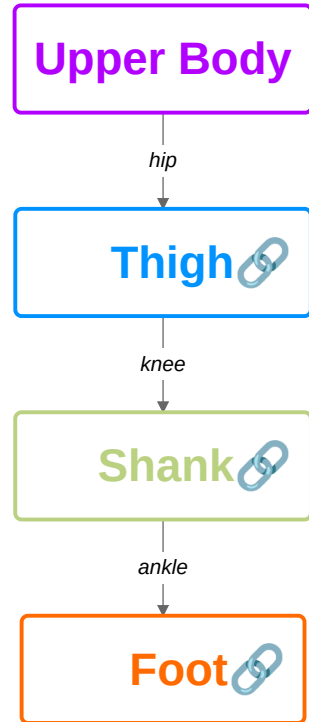




# Lumped Model 🤸



# Mirrored Pitch Joints



# Reduction Layer



# OCP Formulation

$$x(t) = \begin{bmatrix} q(t) \\ \dot{q}(t) \end{bmatrix} \quad (1)$$

$$u(t) = \ddot{q}(t) \quad (2)$$

$$\begin{aligned} f_{expl}(x(t), u(t), t) &= \begin{bmatrix} \dot{q}(t) \\ \ddot{q}(t) \end{bmatrix} \\ &= \dot{x}(t) \end{aligned} \quad (3)$$



# OCP Formulation

$$\min_{x(\cdot), u(\cdot)} \int_0^{T_f} \omega_s l_s(x(t), u(t)) + \omega_h l_h(x(t), t) + \omega_r l_r(u(t)) dt \quad (4)$$

$$\text{s.t.} \quad \begin{bmatrix} q_0 \\ 0 \end{bmatrix} = x(0) \quad (C0)$$

$$\begin{bmatrix} \underline{q}(t) \\ -v_{max} \end{bmatrix} \leq x(t) \leq \begin{bmatrix} \bar{q}(t) \\ v_{max} \end{bmatrix} \quad \forall t \in [0, T_f] \quad (C1)$$

$$-a_{max} \leq u(t) \leq a_{max} \quad \forall t \in [0, T_f] \quad (C2)$$

$$p_{stab}(t) \in \mathbf{P} \quad \forall t \in [0, T_f] \quad (C3)$$

# Lagrange Cost Functions

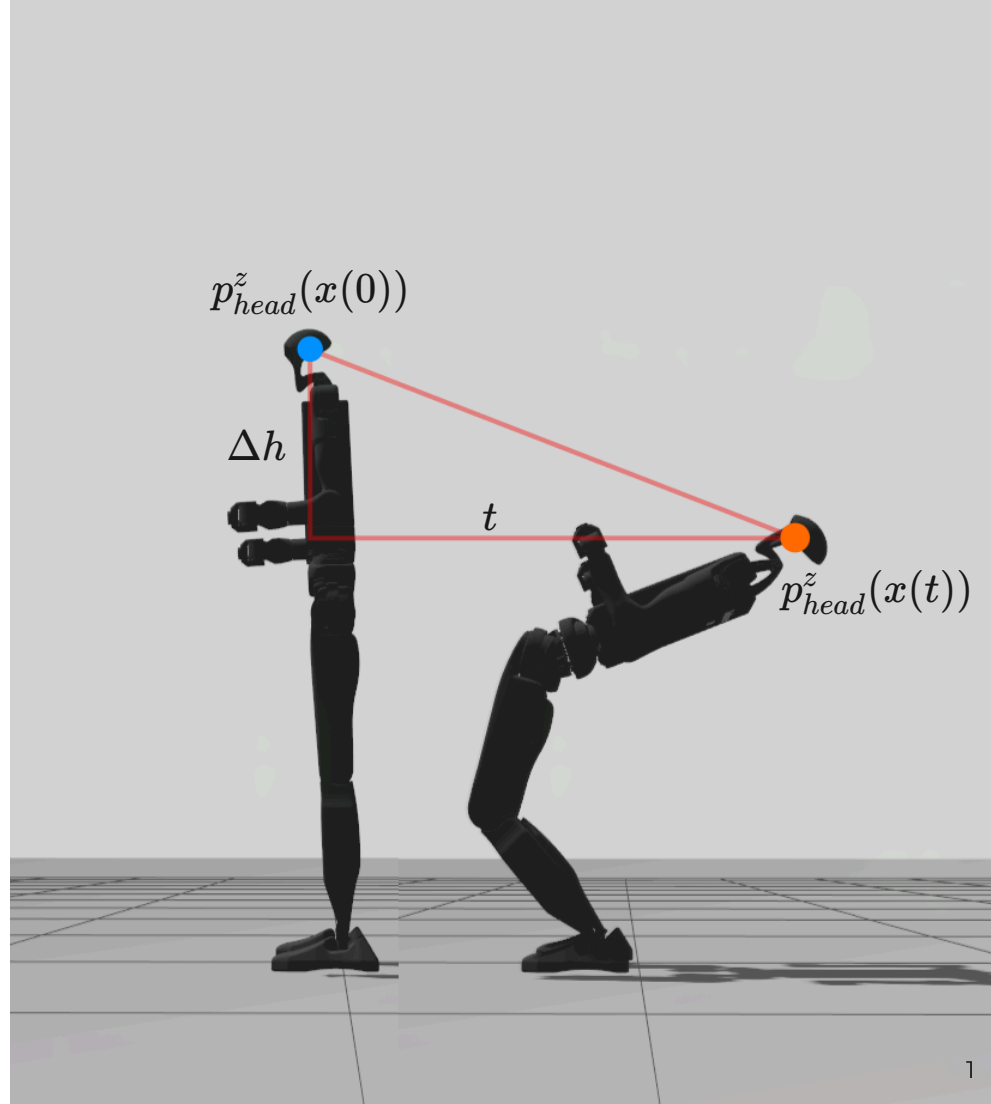
$$l_h(x(t), t) : \text{🎯 Head Motion Tracking} \quad (\text{L0})$$

$$l_s(x(t), u(t)) : \text{⚖️ Stability Metric} \quad (\text{L1})$$

$$l_r(u(t)) = u(t)^2 \quad (\text{L2})$$

# Head Motion Tracking

$$l_h(x(t), t) = (p_{head}^z(x(0)) - \Delta h t - p_{head}^z(x(t)))^2 \quad (\text{L0})$$



# Lagrange Cost Functions

$$l_h(x(t), t) = (p_{head}^z(x(t)) - \Delta h \ t - p_{head}^z(x(0)))^2 \quad (\text{L0})$$

$l_s(x(t), u(t)) :$   Stability Metric

(L1)

$$l_r(u(t)) = u(t)^2 \quad (\text{L2})$$



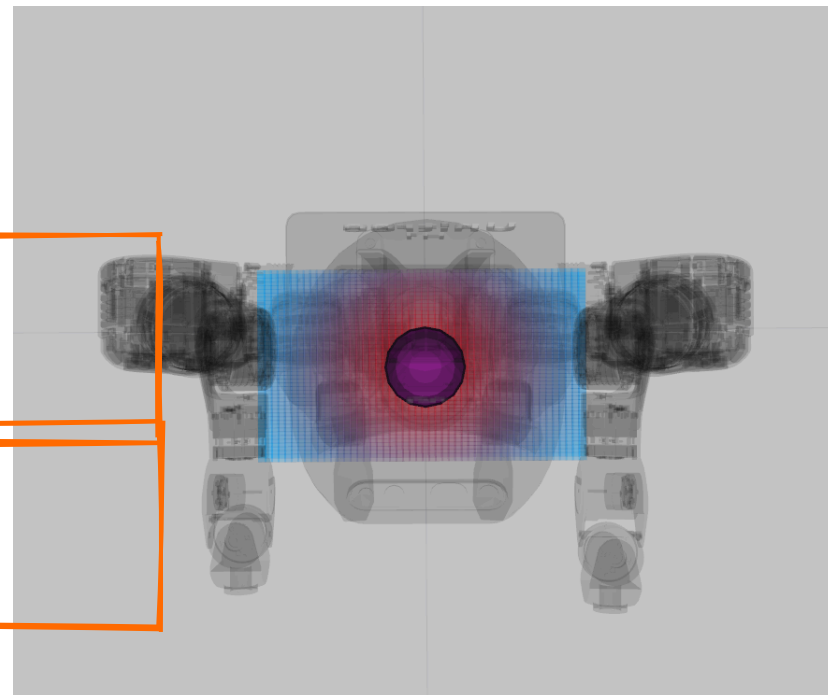
# Previous Criteria

●: low values | ●: high values

 no normalization

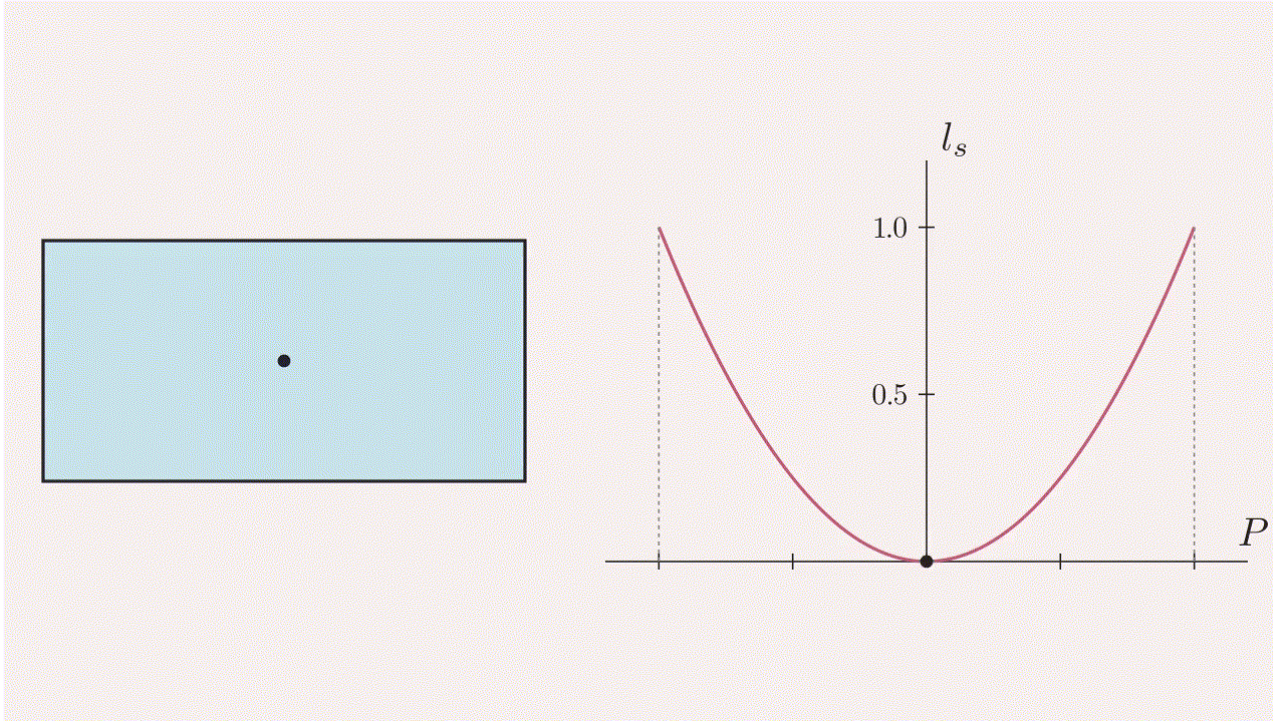
 no limitation to PoS

 projected CoM (*static*)



 Centroid Distance

# Stability Metric



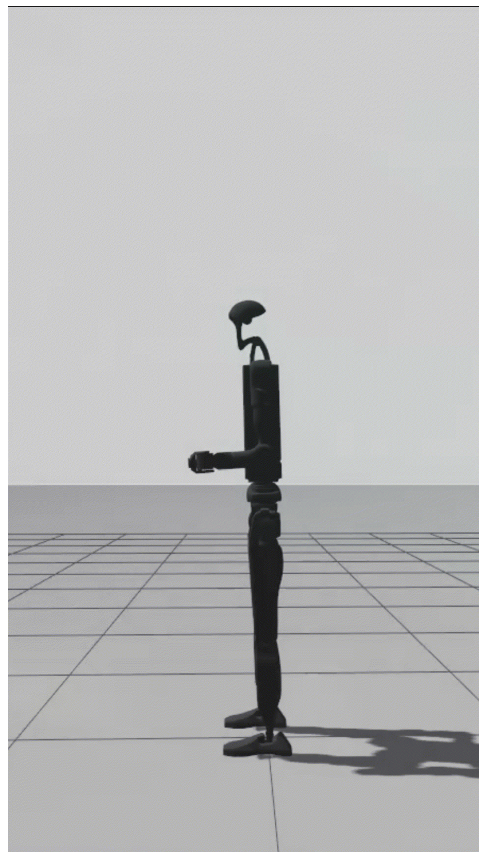
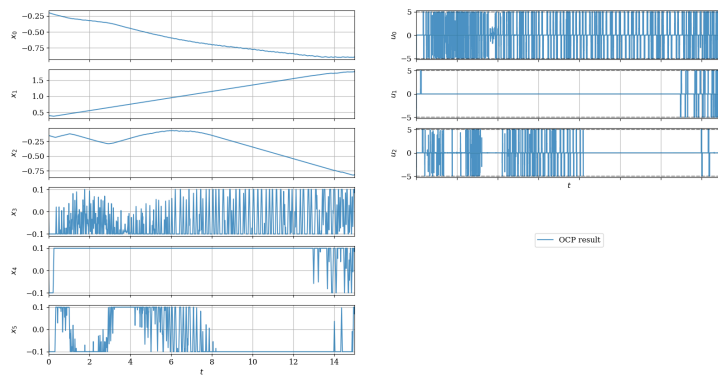
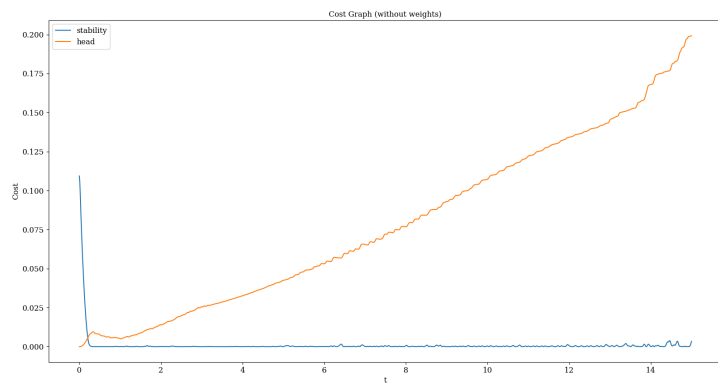
# Lagrange Cost Functions

$$l_h(x(t), t) = (p_{head}^z(x(t)) - \Delta h t - p_{head}^z(x(0)))^2 \quad (\text{L0})$$

$$l_s(x(t), u(t)) = \frac{(p_{stab}(x(t), u(t)) - P_c)^2}{(\underline{P} - P_c)^2} \quad (\text{L1})$$

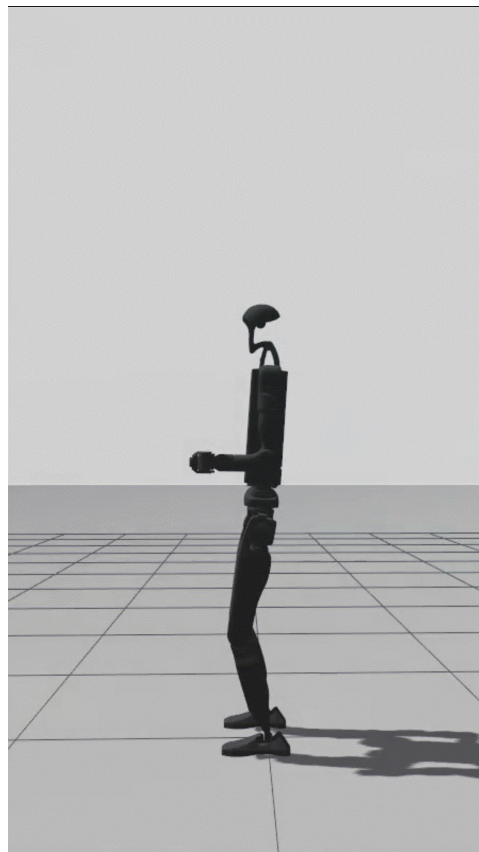
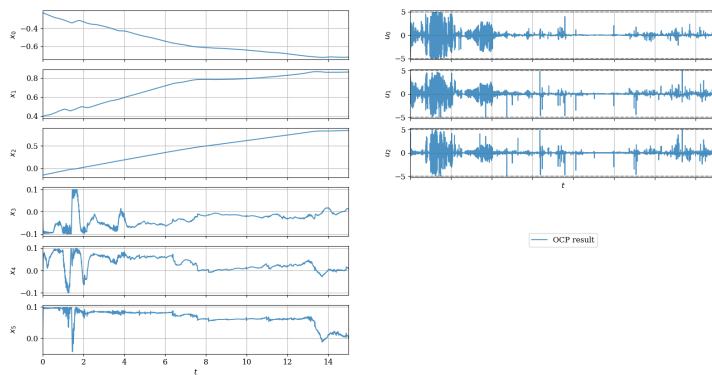
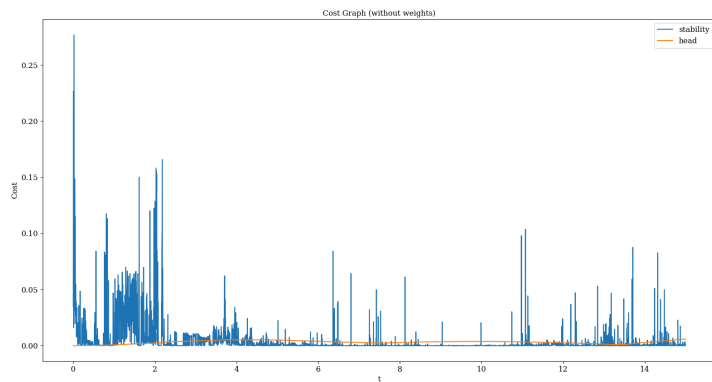
$$l_r(u(t)) = u(t)^2 \quad (\text{L2})$$

# Projected CoM





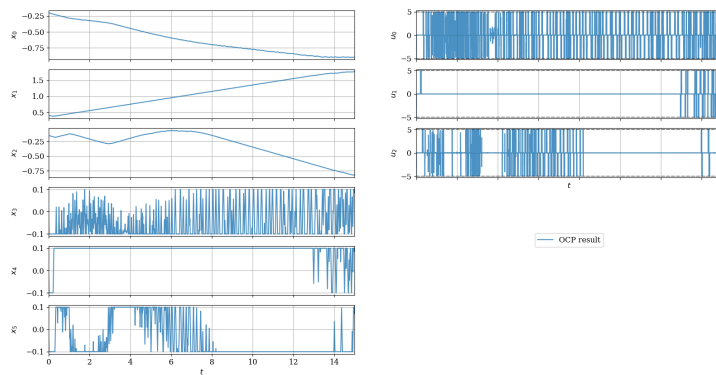
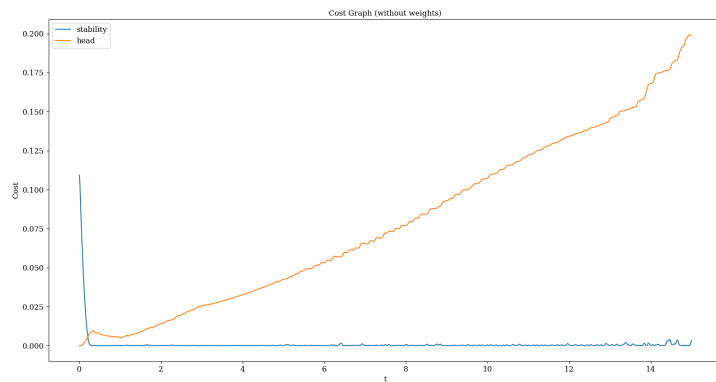
# Zero Moment Point (ZMP)



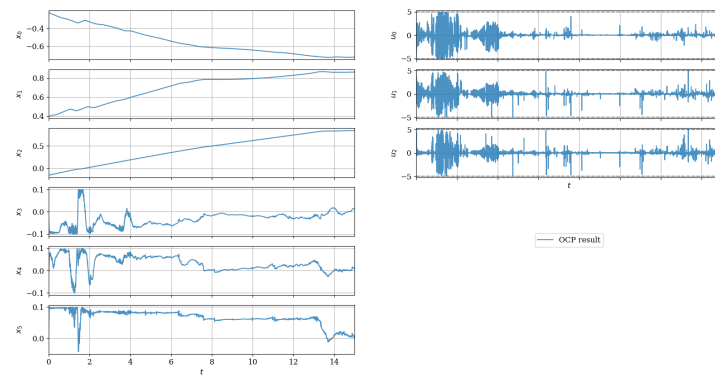
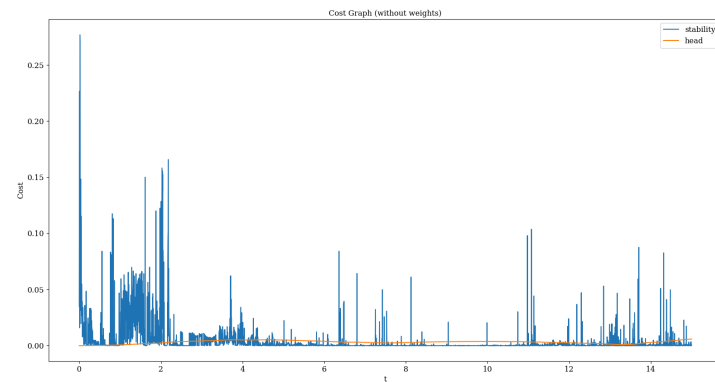




# Projected CoM

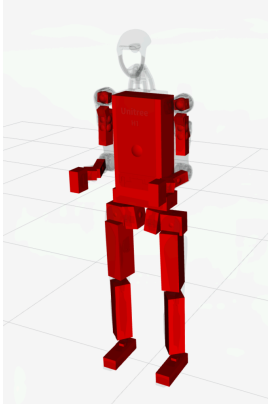


# Zero Moment Point





## Full ZMP



$m_{tot}$  : Total Mass

$m_i$  : Segment Mass

$x, y, z$  : CoM

$g$  : Grav. Acc.

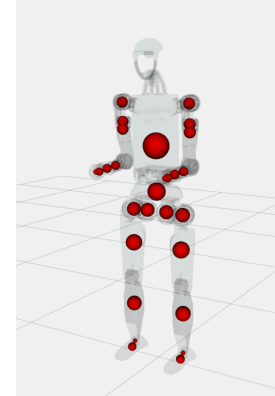
$M_A$  : Angular Momentum

$M_T$  : Translational Momentum

$$p_x = \frac{m_{tot} g x + p_z \dot{M}_T^x - \dot{M}_A^y}{m_{tot} g + \dot{M}_T^z}$$



## Approximated ZMP



$[kg]$

$[kg]$

$[m]$

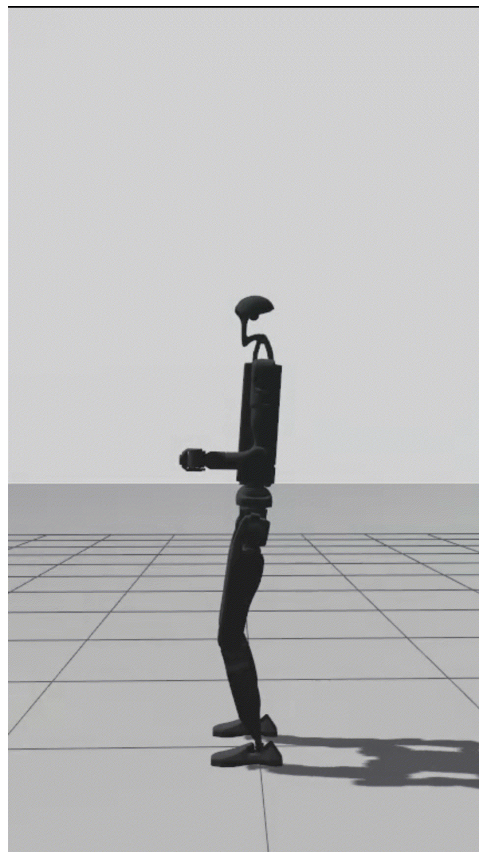
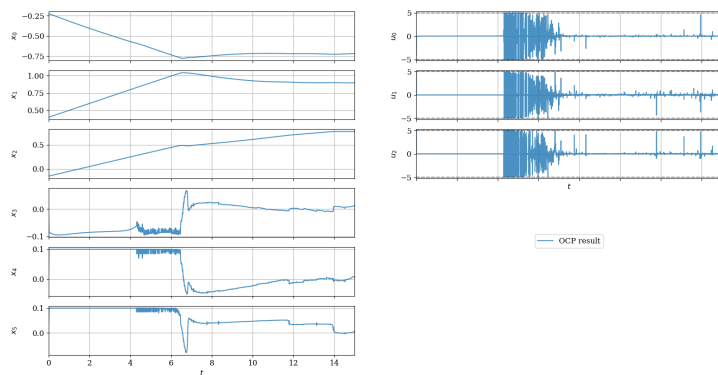
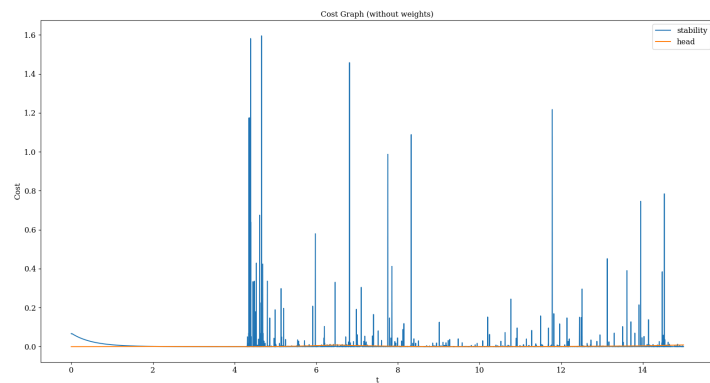
$[m/s^2]$

$[Nm s]$

$[Ns]$

$$p_x = \frac{\sum_{i=0}^N m_i ((\ddot{z}_i + g)x_i - (z_i - p_z)\ddot{x}_i)}{\sum_{i=0}^N m_i (\ddot{z}_i + g)}$$

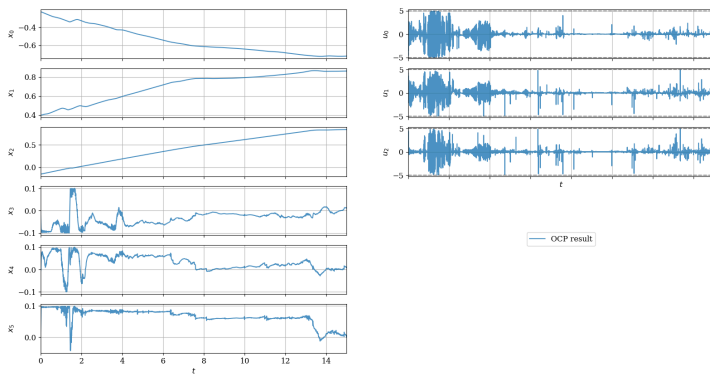
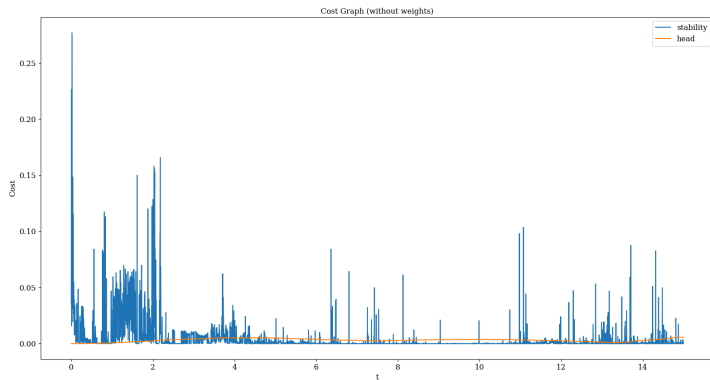
# Approximated ZMP



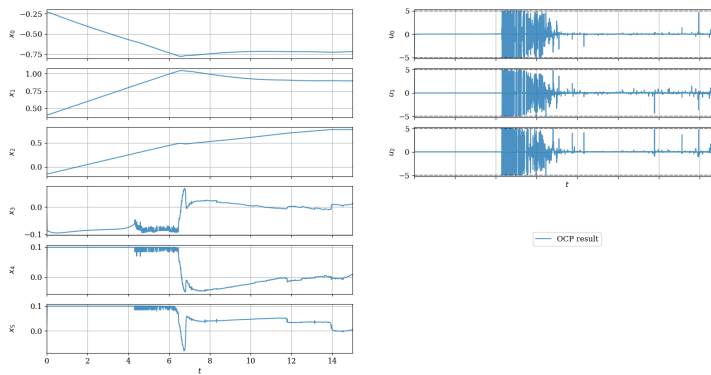
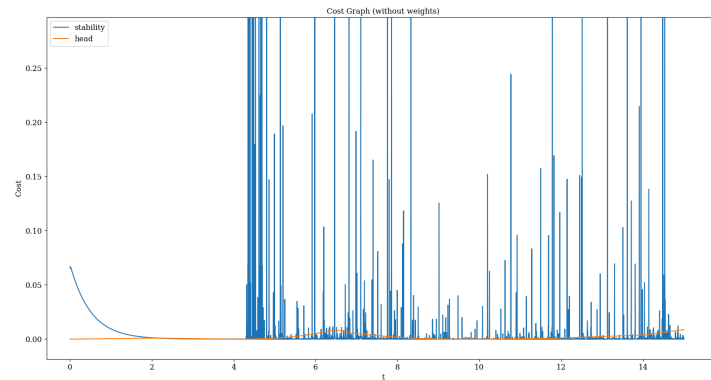




# Full ZMP



# Approximated ZMP





# Future Work



Other Dynamic Stability Criterias



Faster Motion



Sim2Real (real **He1**nz)



Model Predictive Control



Active Dodging

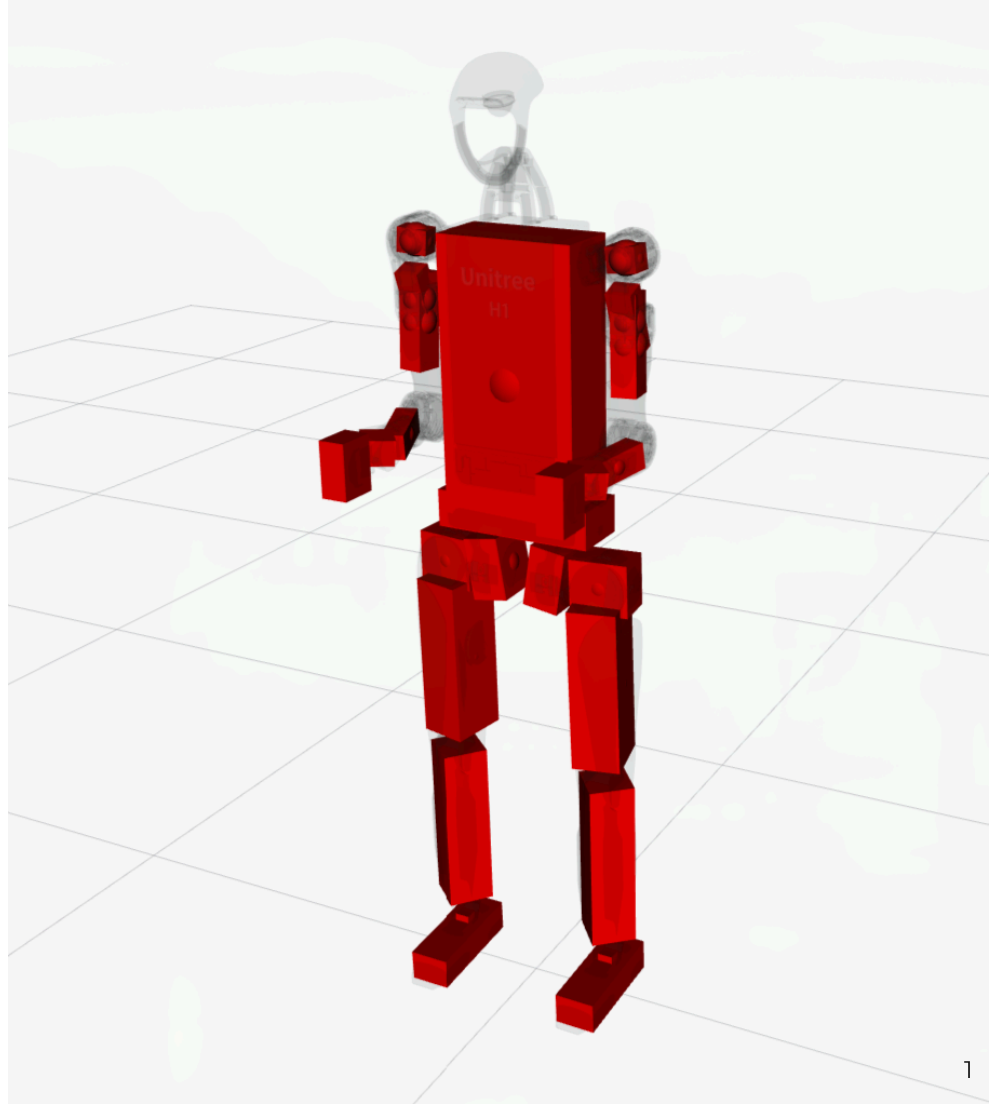
# Ideas for Active Dodging 🚬

Constraint 🚧

$$p_{obs}(t) \notin \text{conv } R(t) \forall t \in [0, T_f]$$

Lagrange Term 💶

$$\frac{1}{d(\text{conv } R(t), p_{obs}(t))} \forall t \in [0, T_f]$$



Thank you for following the White Rabbit.



A 3D rendering of a robotic arm, possibly a Shadow Hand, positioned on a grid floor. The arm is dark and articulated, with its hand open. A mouse cursor is visible in the upper left quadrant of the image. The background is a plain, light gray wall.

# Questions?

# Appendix

## 🤔 Initial Guess



Hip:  
inter.



Knee:  
const.



Ankle:  
const.

